路徑容錯移轉
ONTAP 9

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路徑容錯移轉

路徑容錯移轉總覽

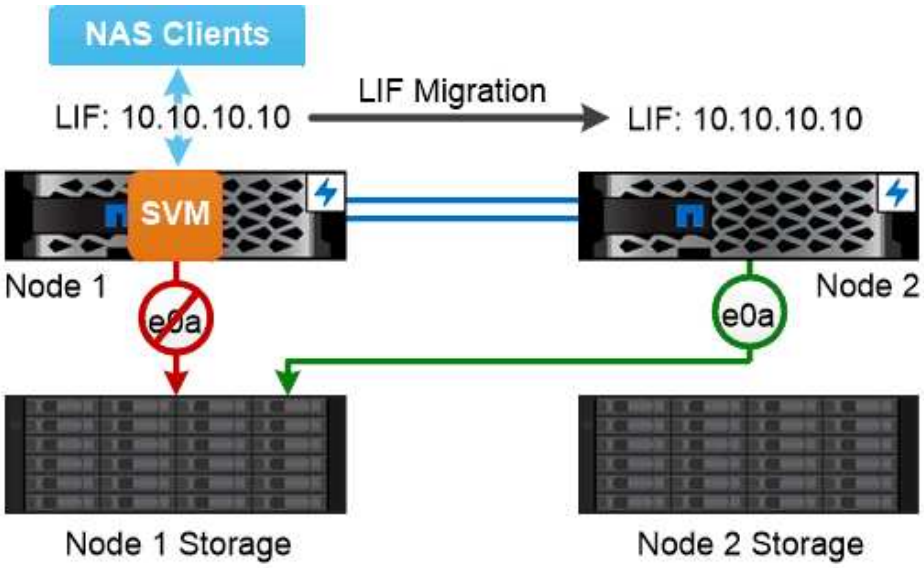
在NAS和SAN拓撲中、由哪種方式進行路徑容錯移轉、有很大的差異ONTAP 。NAS LIF會在連結失敗後自動移轉至不同的網路連接埠。SAN LIF不會移轉（除非您在故障後手動移動）。相反地、主機上的多重路徑技術會將流量轉移到同一個SVM上的不同LIF、但會存取不同的網路連接埠。

NAS路徑容錯移轉

NAS LIF會在目前連接埠的連結失敗後、自動移轉至正常運作的網路連接埠。LIF移轉的連接埠必須是LIF的_Failover群組_成員。容錯移轉群組原則_將資料LIF的容錯移轉目標縮小為擁有資料及其HA合作夥伴的節點上的連接埠。

為了管理上的便利性、ONTAP 在網路架構中為每個_broadcast domain_建立一個容錯移轉群組。廣播網域群組屬於同一個第2層網路的連接埠。例如、如果您使用VLAN來依部門（工程、行銷、財務等）分隔流量、則每個VLAN會定義一個獨立的廣播網域。每次新增或移除廣播網域連接埠時、都會自動更新與廣播網域相關的容錯移轉群組。

使用廣播網域來定義容錯移轉群組、以確保容錯移轉群組保持最新狀態、幾乎永遠都是個好主意。不過、有時候您可能會想要定義與廣播網域無關的容錯移轉群組。例如、您可能希望LIF僅容錯移轉至廣播網域中所定義之連接埠子集內的連接埠。



A NAS LIF automatically migrates to a surviving network port after a link failure on its current port.

子網路_

子網路_保留廣播網域中的IP位址區塊。這些位址屬於同一個第3層網路、當您建立LIF時、會分配給廣播網域中的連接埠。在定義LIF位址時、指定子網路名稱通常比指定IP位址和網路遮罩更容易且更容易出錯。

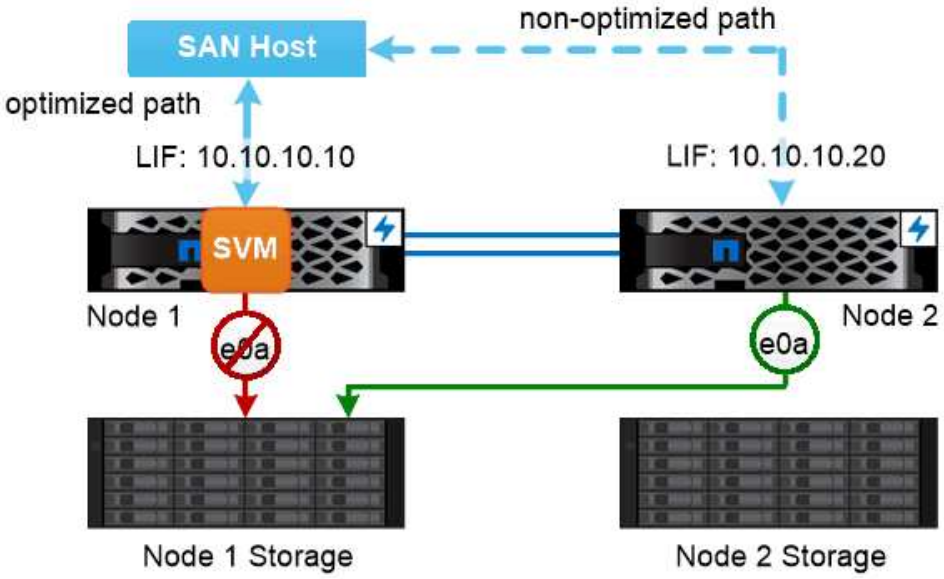
SAN路徑容錯移轉

SAN主機使用ALUA（非對稱邏輯單元存取）和MPIO（多重路徑I/O）、在連結失敗後將流量重新路由至存續的LIF。預先定義的路徑會決定SVM服務的LUN可能路由。

在SAN環境中、主機會被視為向LUN *Target*提出要求的_啟動器。_ MPIO可啟用從啟動器到目標的多個路徑。ALUA會識別最直接的路徑、稱為_最佳化路徑。_

您通常會在LUN所屬節點上設定多個最佳化的生命路徑、並在HA合作夥伴上設定多個未最佳化的生命路徑。如果擁有節點上的某個連接埠故障、主機會將流量路由傳送至正常運作的連接埠。如果所有連接埠都失敗、主機會透過未最佳化的路徑路由流量。

根據預設、「選擇性LUN對應」（SLM）會限制從主機到LUN的路徑數量。ONTAP新建立的LUN只能透過通往擁有LUN或其HA合作夥伴之節點的路徑存取。您也可以在啟動器的連接埠集_中設定lifs、以限制對LUN的存取。



A SAN host uses multipathing technology to reroute traffic to a surviving LIF after a link failure.

在SAN環境中移動磁碟區_*

根據預設、ONTAP 將從SAN主機到LUN的路徑數量限制為由SURE_Selective LUN Map（SLM__）。新建立的*LUN*只能透過通往擁有*LUN*或其*HA*合作夥伴（*LUN*的_報告節點）節點的路徑存取。

這表示當您將磁碟區移至另一個HA配對上的節點時、您需要將目的地HA配對的報告節點新增至LUN對應。然後您可以在MPIO設定中指定新的路徑。磁碟區移動完成後、您可以從對應中刪除來源HA配對的報告節點。

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